

## Answers to Quiz 8: Simpson's Rule

(Numerical Analysis)

1. (2 points) Three parabolas need to be drawn. The first parabola passes through the points

$$(0, f(0)), \quad (40, f(40)), \quad (80, f(80)).$$

The second parabola passes through

$$(80, f(80)), \quad (120, f(120)), \quad (160, f(160)).$$

The third parabola passes through

$$(160, f(160)), \quad (200, f(200)), \quad (240, f(240)).$$

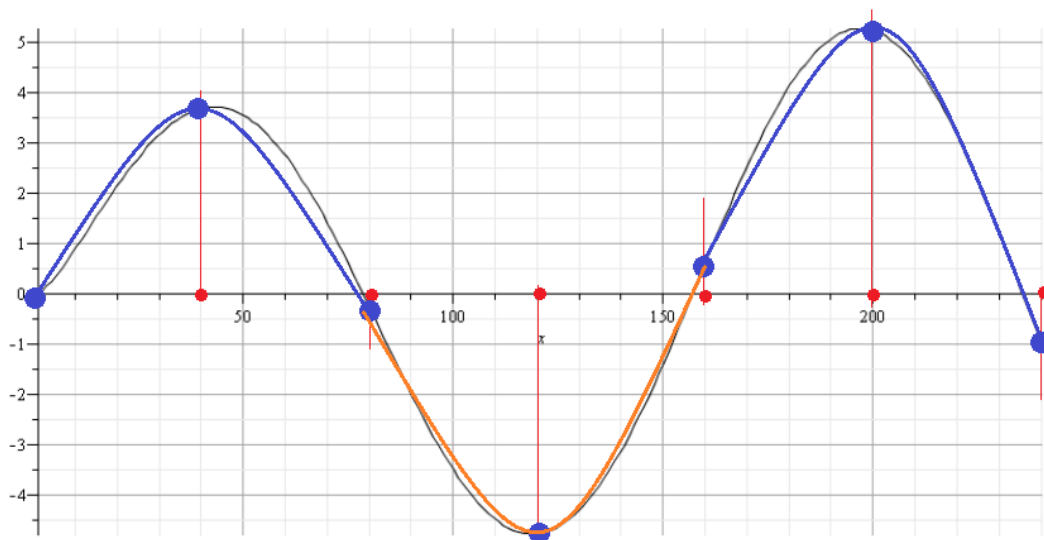


Figure 1: Sketch parabolas here.

2. (5 points) We want to approximate the integral

$$I = \int_2^4 \frac{dx}{\ln x}.$$

Let

$$f(x) = \frac{1}{\ln x}.$$

The interval of integration is

$$[a, b] = [2, 4].$$

We are given a uniform lattice consisting of 9 points. Therefore, the interval is divided into

$$n = 9 - 1 = 8$$

equal subintervals.

This is suitable for Simpson's rule, since the number of subintervals must be even.

For a uniform lattice, the step size is

$$h = \frac{b - a}{n}.$$

Hence,

$$h = \frac{4 - 2}{8} = \frac{2}{8} = 0.25.$$

Thus, the grid points are given by

$$x_i = a + ih = 2 + 0.25i, \quad i = 0, 1, \dots, 8.$$

Therefore,

$$\begin{aligned} x_0 &= 2, \\ x_1 &= 2.25, \\ x_2 &= 2.50, \\ x_3 &= 2.75, \\ x_4 &= 3.00, \\ x_5 &= 3.25, \\ x_6 &= 3.50, \\ x_7 &= 3.75, \\ x_8 &= 4.00. \end{aligned}$$

We now calculate

$$f(x_i) = \frac{1}{\ln x_i}$$

at each grid point.

| $i$ | $x_i$ | $f(x_i) = \frac{1}{\ln x_i}$ |
|-----|-------|------------------------------|
| 0   | 2.00  | 1.442695                     |
| 1   | 2.25  | 1.233152                     |
| 2   | 2.50  | 1.091357                     |
| 3   | 2.75  | 0.988532                     |
| 4   | 3.00  | 0.910239                     |
| 5   | 3.25  | 0.848425                     |
| 6   | 3.50  | 0.798236                     |
| 7   | 3.75  | 0.756569                     |
| 8   | 4.00  | 0.721348                     |

For an even number  $n$  of subintervals, the composite Simpson's rule is

$$\int_a^b f(x) dx \approx \frac{h}{3} [f(x_0) + 4f(x_1) + 2f(x_2) + 4f(x_3) + \cdots + 2f(x_{n-2}) + 4f(x_{n-1}) + f(x_n)].$$

The first and last points have coefficient 1. The interior points have alternating coefficients 4 and 2:

$$1, 4, 2, 4, 2, 4, 2, 4, 1.$$

Since  $n = 8$ , Simpson's rule gives

$$\begin{aligned} I \approx \frac{h}{3} [ & f(x_0) + 4f(x_1) + 2f(x_2) + 4f(x_3) \\ & + 2f(x_4) + 4f(x_5) + 2f(x_6) + 4f(x_7) + f(x_8) ]. \end{aligned}$$

Since

$$h = 0.25,$$

we obtain

$$\begin{aligned} I \approx \frac{0.25}{3} [ & 1.442695 + 4(1.233152) + 2(1.091357) + 4(0.988532) \\ & + 2(0.910239) + 4(0.848425) + 2(0.798236) \\ & + 4(0.756569) + 0.721348 ]. \end{aligned}$$

Calculating the expression inside the brackets gives

$$I \approx \frac{0.25}{3} [1.442695 + 4.932608 + 2.182714 + 3.954128 \\ + 1.820478 + 3.393700 + 1.596472 + 3.026276 + 0.721348].$$

Therefore,

$$I \approx \frac{0.25}{3} (23.070417).$$

Finally,

$$\int_2^4 \frac{dx}{\ln x} \approx 1.922535.$$

Thus, using Simpson's rule with a uniform lattice of 9 points, the approximate value of the integral is

$$I \approx 1.9225.$$